

TODO: Paper Title

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Abstract

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CCS Concepts

• Security and privacy → Domain-specific security and privacy architectures.

Keywords

Audio Large Language Models, Adversarial Attack, Emotion Recognition, Speech Security

1 Observation

1.1 Clean emotion recognition is governed by structured non-target preference

We first examine whether clean emotion recognition in the model behaves as a hard one-hot decision, or instead as a structured competition over candidate emotions. The latter is clearly supported by our results. Under clean speech, the model does not place all of its confidence on the ground-truth emotion; rather, it consistently preserves non-trivial mass on a small set of emotionally plausible alternatives. Crucially, this residual mass is not random. It follows stable, class-dependent preference patterns, indicating that the model’s prediction behavior is governed by a biased geometry within the emotion subspace rather than by generic uncertainty over the whole vocabulary.

Figure ?? shows this phenomenon at the candidate-posterior level. None of the four classes is close to one-hot. Angry and sad show only moderate dominance, with candidate-set accuracies of 0.61 and 0.63, while happy and neutral are much less stable, with accuracies of 0.15 and 0.24. More importantly, the strongest competitor is not uniform across classes. Sad is the main competitor for true angry, whereas angry is the main competitor for true happy, neutral, and sad. The same asymmetry appears at the prototype level: the closest pair is angry–happy (JS divergence 0.0195), followed by angry–neutral (0.0283), showing that the clean emotion space is already unevenly entangled before any attack is introduced.

Figure ?? provides token-level evidence for the same claim. At the first decoding step, unrelated vocabulary tokens remain strongly suppressed across all four classes, with mean logits around -2.3

Table 1: Response quality degradation under emotion misperception (Δ = Aligned–Conflict; higher = greater degradation). 1–5 Likert scale; $n=4$ per emotion, $N=20$.

Emotion	Degradation (Δ)		
	$\Delta_{\text{Faith.}}$	$\Delta_{\text{Emp.}}$	$\Delta_{\text{Rel.}}$
Angry	+0.75	0.00	0.00
Happy	+1.75	+3.00	+2.50
Sad	0.00	0.00	0.00
Neutral	+1.50	+2.00	+0.75
Surprise	+2.50	+1.00	+2.00
Overall	+1.30	+1.20	+1.05

to -2.5 , while specific non-target emotion tokens remain highly activated. For example, on true happy samples, the target token happy has mean first-step logit 11.20, lower than both angry (14.48) and neutral (13.39); on true neutral samples, the target token neutral has mean logit 13.44, but both angry (14.34) and sad (14.14) are higher. This contrast is important: the model is not broadly uncertain over the full vocabulary. Instead, its uncertainty is concentrated inside the emotion token set, where certain alternative emotions are systematically preferred over others. Taken together, the posterior-level and logit-level evidence support a single observation: clean emotion recognition in the model is governed by structured non-target preference. This suggests that downstream objectives should account for inter-emotion geometry, rather than assuming that the clean model already represents emotion as a purely one-hot target.

1.2 Downstream Impact of Emotion Misperception

Beyond the logit-level vulnerability established in Section ??, we ask whether emotion misperception propagates into the model’s downstream response behavior. We design an attack-agnostic controlled experiment: for each audio sample, the model (Voxtral-Mini-3B [?]) receives an *Aligned* prompt (truthful emotion description) and a *Conflict* prompt (false emotion with maximum affective contrast). We evaluate $N = 20$ samples from ESD [?] across five emotions ($n=4$ per emotion). An independent LLM judge (DeepSeek V3 [?]) rates each response on Faithfulness, Empathy, and Relevance (1–5 Likert). We note that this is a controlled pilot study intended to demonstrate the *existence* and *direction* of degradation effects rather than to estimate precise effect sizes.

(1) Misperception causes systematic response degradation with pronounced emotion asymmetry. As shown in Table ??, the Conflict condition degrades the overall average across all three dimensions ($\Delta_{\text{Faith.}} = +1.30$, $\Delta_{\text{Emp.}} = +1.20$, $\Delta_{\text{Rel.}} = +1.05$), but

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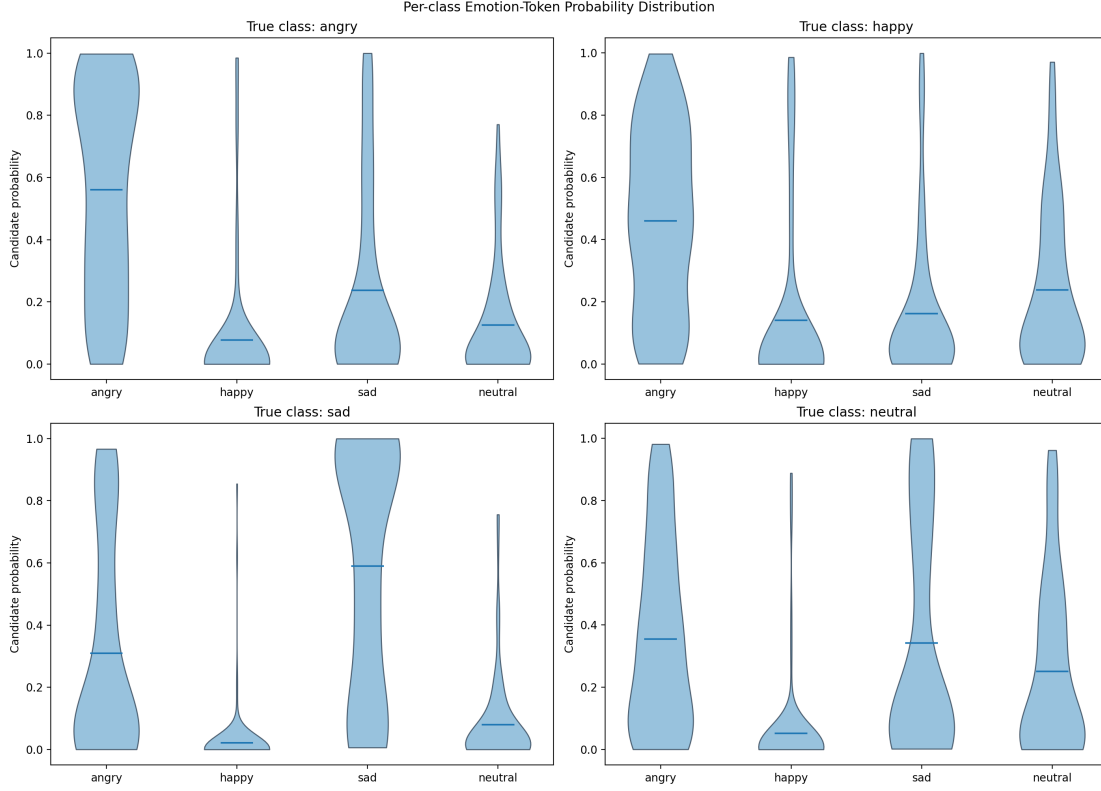


Figure 1: Per-class candidate probability distributions over angry, happy, sad, and neutral under clean audio-only inputs. The posteriors are not one-hot. Each true class retains substantial probability mass on specific competing emotions, revealing structured non-target preference rather than uniform uncertainty.

the effect is highly asymmetric across source emotions. *Happy* is the most vulnerable ($\Delta_{\text{Emp.}} = +3.00$, $\Delta_{\text{Rel.}} = +2.50$), while *sad* is completely robust ($\Delta = 0$ across all dimensions)—the model even explicitly overrides the misleading prompt in some cases, stating “*They are not angry or frustrated, but rather expressing their grief and pain.*” This pronounced emotion-dependent asymmetry—where *happy* and *surprised* show the largest degradation while *sad* and *angry* remain relatively stable—is consistent with the logit-level fragility of *happy* identified in Section ??.

(2) Misperception induces hallucination in emotionally ambiguous inputs. Neutral audio exhibits a distinct failure mode: the model fabricates emotional narratives with no basis in the audio. For a factual statement (“*The package is scheduled to arrive tomorrow afternoon*”) paired with an anger prompt, the model attributes frustration to the speaker, constructs causal explanations, and generates coping strategies—none grounded in the audio content (Faithfulness drops to 1.50/5.00). This indicates that when audio carries weak emotional signals, the model defaults to the text prompt as its primary emotion source, making emotionally neutral speech—arguably the most frequently encountered category in practical deployments—a particularly vulnerable attack surface.

1.3 Insufficiency of Traditional SER Attack Methods

We evaluate whether adversarial attack methods developed for Speech Emotion Recognition (SER) transfer to ALLMs. We adapt four representative prior works—spanning gradient-based, generator-based, and distortion-based paradigms—yielding six concrete attack instantiations against Voxtral-Mini-3B. All attacks are *untargeted* (strictly easier than targeted flipping). For gradient-based and generator-based methods, we grant full white-box access with a perturbation budget of $\epsilon = 0.03$ (L_∞), which is $3.75\times$ larger than our ALLM-native attacks ($\epsilon = 0.008$). The four prior works are:

- **STAA-Net** [?]: A generator-based attack using a Wave-U-Net generator with C&W untargeted loss [?] and sparsity mask. Trained end-to-end with white-box access to Voxtral (200 train / 200 test samples from ESD).
- **PGD** [?]: Projected Gradient Descent on a surrogate SER classifier (wav2vec2-based), achieving 100% attack success rate on the surrogate, then transferred to Voxtral (525 test samples).
- **Gao et al.** [?]: Three black-box speech distortions—Vocal Tract Length Normalization (VTLN), McAdams coefficient

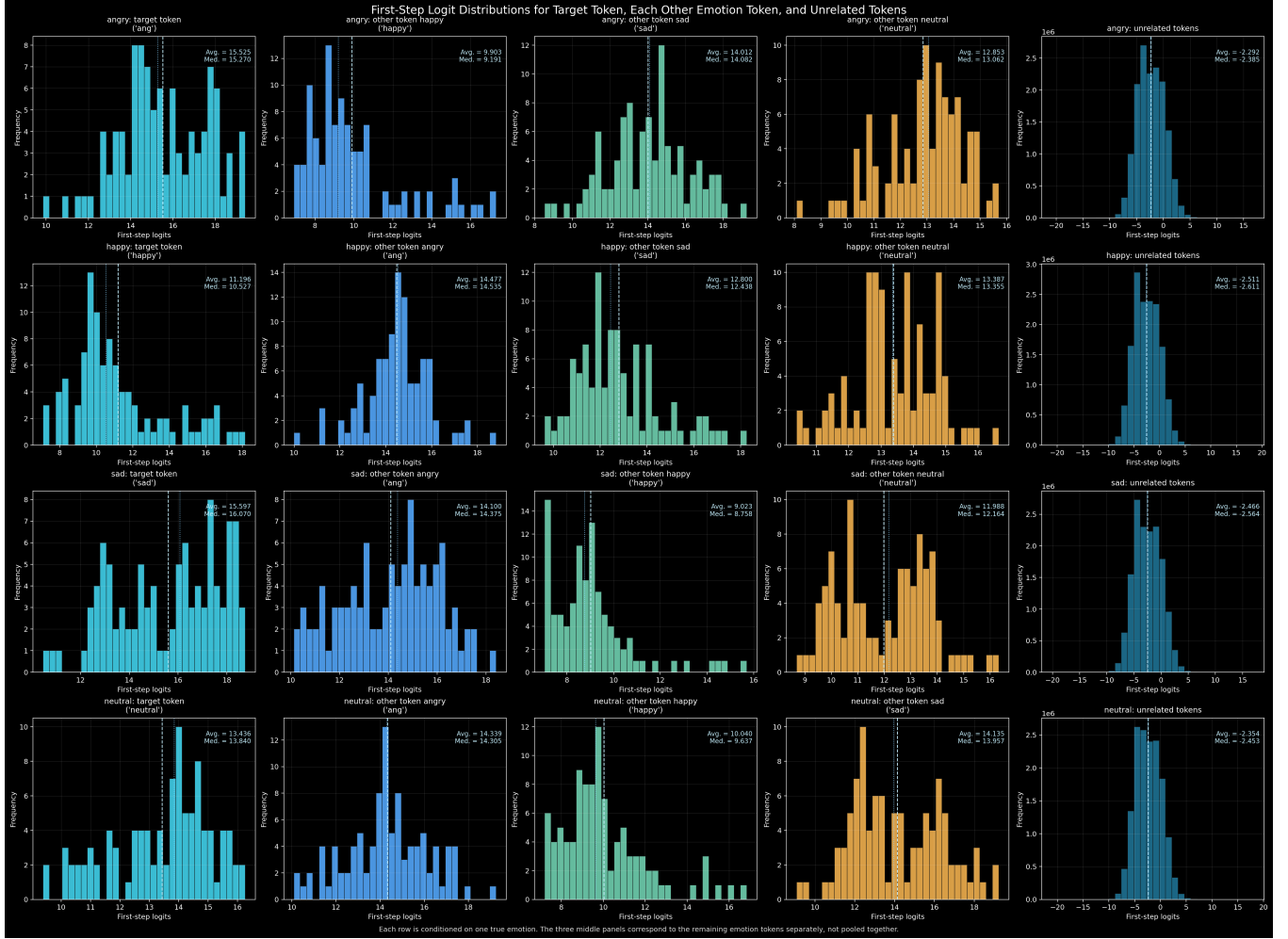


Figure 2: First-step logit distributions for the target emotion token, each non-target emotion token, and unrelated tokens. Unrelated tokens remain strongly suppressed across all classes, whereas specific non-target emotion tokens consistently receive elevated logits. This shows that the model’s uncertainty is concentrated within the emotion subspace, where certain competing emotions are systematically preferred over others.

transformation, and Modulation Spectrum Smoothing (MSS)—originally proposed for SER robustness evaluation (525 test samples each).

- **Ren et al. [?]:** An atrous CNN generator trained via lifelong learning to attack SER models, then transferred to Voxtral (525 test samples).

Figure ?? reports the ALLM prediction change rate—the fraction of test samples whose Voxtral emotion prediction differs between clean and adversarial audio. All six attack instantiations fail: prediction change rates range from 2.5% (STAA-Net) to 10.5% (PGD), with distortion-based methods at 4.2–9.3% and generator-based methods at 2.5–2.9%. The failure is particularly striking for PGD, which achieves 100% attack success on its surrogate SER classifier yet alters only 10.5% of Voxtral predictions. For reference, our ALLM-native method (Section ??) achieves 93.8% *targeted* attack success rate with a 3.75× smaller perturbation budget. We note

that the comparison is directional rather than metric-equivalent, as the baselines perform untargeted disruption while our method performs strictly harder targeted flipping.

The consistent failure across all paradigms suggests a fundamental architectural incompatibility rather than insufficient perturbation magnitude. SER attacks optimize perturbations against standalone classifiers with fixed output spaces, whereas ALLMs route audio through an encoder–adapter–LLM pipeline where emotion judgments emerge via autoregressive generation over a vocabulary of tens of thousands of tokens. We hypothesize that the perturbation signal, shaped for a classifier’s compact decision boundary, is attenuated as it passes through the LLM’s billion-parameter processing layers before reaching the emotion-relevant decoding steps. Regardless of the precise mechanism, the empirical outcome is clear: existing SER attack paradigms do not transfer to ALLMs, motivating the ALLM-native methodology presented in Section ??.

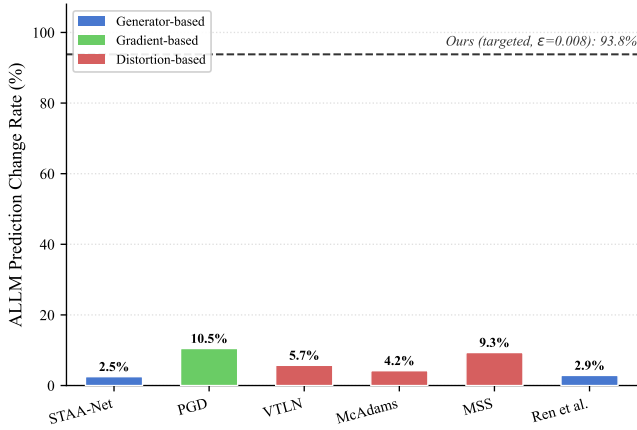


Figure 3: ALLM prediction change rate under six SER adversarial attack methods transferred to Voxtral-Mini-3B. All methods alter fewer than 10.5% of predictions despite using untargeted attacks with 3.75× larger perturbation budgets. The dashed line indicates the targeted attack success rate of our ALLM-native method (Section ??).

2 Threat Model

As emotion-aware ALLMs enter deployment in affective companion systems, mental health support, and intelligent customer service, their downstream responses become conditioned on an internal emotion percept—the model’s implicit interpretation of the speaker’s emotional state—that the user can neither observe nor correct. Section ?? demonstrated that even a single misperceived emotion is enough to trigger systematic response degradation—faithfulness collapse on neutral inputs, hallucinated emotional narratives, and a pronounced vulnerability gap between positive and negative valences. This fragility motivates a concrete adversarial question: can an attacker reliably steer the model’s emotion percept to a chosen target, using only an imperceptible audio perturbation?

Attacker’s goal. Given clean audio x with ground-truth speaker emotion e_s , the attacker seeks a perturbation δ such that the adversarial audio $x' = x + \delta$ causes the target ALLM to output a pre-specified emotion $e_t \neq e_s$, while the linguistic content and perceptual quality of the waveform remain intact. The attack is *targeted*: the attacker controls not only *whether* the prediction changes but *which* emotion it changes to—a strictly harder objective than untargeted disruption.

Attacker’s knowledge. We operate in a white-box setting: the attacker has full access to the target ALLM’s architecture, weights, and intermediate representations, and can compute end-to-end gradients through the entire audio-encoder-adapter-LLM pipeline. The attacker knows the prompt templates used for emotion elicitation but cannot modify model parameters or prompts; only the input waveform is under adversarial control. We assume the adversarial audio is injected as a digital input (e.g., uploaded file or API-mediated stream) rather than played over-the-air, so channel distortions are not modeled.

Constraints. The perturbation must satisfy three requirements. An L_∞ norm bound $\|\delta\|_\infty \leq \epsilon$ with $\epsilon = 0.008$ enforces bounded sample-level waveform perturbation as a hard constraint, applied by projection after each optimization step. Semantic preservation requires that the model’s own transcription of the adversarial audio remains highly similar to that of the clean audio, ensuring the linguistic content is unaltered. A multi-resolution STFT magnitude loss further penalizes perceptually salient spectral distortions. The latter two are implemented as soft loss terms whose formulations are given in Section ??.

Formal problem statement. Let f denote the target ALLM, p_{emo} an emotion elicitation prompt, and p_{asr} a transcription prompt. Given clean audio $x \in \mathbb{R}^T$ with source emotion e_s and an attacker-chosen target $e_t \neq e_s$, the objective is to find δ such that the adversarial audio $x' = x + \delta$ satisfies:

- (i) $f(x', p_{\text{emo}}) = e_t$ (targeted emotion flip)
- (ii) $\text{sim}(\text{ASR}_f(x', p_{\text{asr}}), \text{ASR}_f(x, p_{\text{asr}})) \geq \tau$ (semantic preservation)
- (iii) $\|\delta\|_\infty \leq \epsilon$ (imperceptibility)

where $\text{ASR}_f(\cdot, p_{\text{asr}})$ denotes the model’s transcription output and $\text{sim}(\cdot, \cdot)$ is a semantic similarity measure detailed in Section ??.

3 Experiments

3.1 Setup

Target models. We evaluate the proposed attack on two architecturally distinct ALLMs. **Voxtral-Mini-3B** [?] follows the prevalent encoder-adapter-LLM pipeline: a Whisper-based audio encoder extracts frame-level features, a linear adapter projects them into the embedding space of a 3B-parameter Mistral backbone, and the LLM generates responses autoregressively. **OpenS2S** [?] takes a different route: a dedicated speech understanding module produces a fused representation tensor that jointly encodes semantic and paralinguistic information, which a Qwen-based LLM then consumes for response generation. The two models differ not only in parameter count but in how emotional information is routed internally—Voxtral distributes emotion across a shared encoder-adapter pathway, while OpenS2S concentrates it in a specialized perception module. This architectural contrast makes the pair a strong test of whether our attack generalizes beyond a single pipeline design.

Dataset and evaluation protocol. All experiments use the Emotional Speech Dataset (ESD) [?], covering four source emotions (angry, neutral, sad, surprise) each paired with three distinct target emotions. For Voxtral we attack $N = 2,000$ samples from 20 speakers across both English and Chinese subsets; for OpenS2S we scale to $N = 8,949$ samples from 9 speakers. Each adversarial audio is evaluated under three semantically varied emotion elicitation prompts, and the final prediction is determined by majority vote—a sample counts as a targeted success only if at least two of three prompts yield the attacker-specified target e_t .

Semantic preservation is assessed by two complementary metrics. The first is the cosine similarity between sentence embeddings of the model’s clean and adversarial transcriptions, thresholded to yield a binary preservation label. The second, available for OpenS2S, is an independent LLM judge (DeepSeek V3 [?]) that determines

Table 2: White-box attack results on two architecturally distinct ALLMs. ASR is determined by majority vote over three emotion prompts. Semantic preservation uses cosine-similarity thresholding for Voxtral and LLM-judge evaluation for OpenS2S. Joint success requires both targeted emotion flipping and semantic preservation.

	Voxtral-Mini-3B	OpenS2S
Samples / Speakers	2,000 / 20	8,949 / 9
Overall ASR	93.80%	78.44%
Best single-prompt ASR	99.90%	94.66%
Semantic preservation	39.75%	19.43%
Joint success	36.40%	15.07%
SNR (dB)	20.60	16.43
$\ \delta\ _\infty$	0.008	0.008

whether the adversarial transcription preserves the speaker’s intent, factual content, and emotional attitude; the two metrics agree on 90.60% of samples. *Joint success* requires both targeted emotion flipping and semantic preservation.

Attack configuration. The perturbation budget is $\varepsilon = 0.008 (L_\infty)$. We use the two-stage optimization schedule described in Section ?? : Stage A runs 20 steps with emotion-dominant loss weighting, followed by Stage B with 40 steps of balanced weighting, for a total of 60 optimization steps per sample. Multi-prompt ensemble averages the emotion loss across all three prompts at each step, and Expectation over Transformation (EoT) applies random time-shifts and gain perturbations to improve robustness to input variation.

3.2 Targeted Emotion Flipping

Table ?? summarizes the headline numbers. On Voxtral, the attack achieves a targeted attack success rate (ASR) of **93.80%** via majority vote, with individual prompts reaching up to 99.90%. On OpenS2S, the ASR is **78.44%**, with the strongest single prompt at 94.66%. Both figures are obtained under an L_∞ budget of 0.008—3.75× smaller than the budget granted to the SER baseline methods evaluated in Section ??, none of which exceeded a 10.5% *untargeted* prediction change rate. This gap underscores a key finding: a perturbation budget that leaves traditional SER attacks essentially inert suffices for near-complete targeted emotion control when the attack is designed for the autoregressive pipeline.

The 15.36-point ASR gap between the two models is instructive. It does not simply reflect model size—both operate in the 3B–7B range—but likely relates to how each architecture represents emotion internally. We conjecture that Voxtral distributes emotion perception across a shared encoder–adapter pathway that was not explicitly trained for SER, yielding relatively diffuse internal emotion representations that can be redirected by modest gradient pressure. OpenS2S, by contrast, channels emotion through a dedicated speech understanding module that produces a more compact, emotion-concentrated representation, potentially creating a more rigid decision boundary. The convergence statistics are consistent with this hypothesis: 99.90% of Voxtral samples converge (emotion loss < 1.0) within the 60-step budget at a mean of 5.4 steps, whereas

Table 3: Per-emotion targeted ASR (% , majority vote). Source emotion indicates the ground-truth label of the clean audio; the attack steers the prediction toward a different, attacker-chosen target.

Source emotion	Voxtral	OpenS2S
Angry	92.20	71.21
Neutral	92.00	81.10
Sad	96.00	81.54
Surprise	95.00	80.58
Overall	93.80	78.44

only 47.85% of OpenS2S samples reach the same threshold, with those that do requiring a mean of 39.5 steps. We note that dataset differences (language mix, sample count, speaker pool) may also contribute to the observed gap, and a matched-subset comparison would be needed to fully isolate the architectural factor.

Per-emotion analysis. Table ?? breaks down the ASR by source emotion. On Voxtral, all four emotions exceed 92%, with sad (96.00%) and surprise (95.00%) being the easiest to flip. On OpenS2S, a wider spread emerges: angry is the hardest source at 71.21%, while sad and neutral both exceed 81%. The relative resistance of angry on OpenS2S resonates with the observation in Section ?? that angry acts as the dominant logit competitor across most other emotion classes: its internal activation is already high, so the model’s confidence in angry is structurally anchored and harder to displace. Conversely, sad—which showed the highest clean-audio dominance for its own class in Section ??—proves easy to redirect, possibly because the model’s concentrated confidence provides a clear gradient target for the optimizer to exploit. We note an apparent tension with Section ??, where sad was robust to prompt-level emotion conflict; this suggests that robustness to textual misdirection does not imply robustness to acoustic-level adversarial perturbation, as the two attack surfaces engage different processing pathways.

3.3 Semantic Preservation and Perturbation Quality

Flipping the model’s emotion percept is only half the objective; a practical attack must also leave the linguistic content of the utterance intact. Table ?? reveals that semantic preservation is the primary bottleneck: joint success rates stand at 36.40% for Voxtral and 15.07% for OpenS2S—roughly one-third of the corresponding ASRs.

On Voxtral, the mean cosine similarity between clean and adversarial transcriptions is 0.6448 ($\sigma = 0.3050$), and 15.25% of samples achieve perfectly identical transcriptions (WER = 0). On OpenS2S, the LLM judge rates 19.43% of adversarial transcriptions as semantically preserved, closely tracking the cosine-similarity-based estimate of 20.96%. The gap between models can be partly attributed to convergence speed: Voxtral samples that converge in a handful of steps experience less optimization pressure on the speech-content features, whereas the prolonged optimization required for OpenS2S gives the perturbation more opportunity to drift into content-bearing acoustic dimensions.

Perturbation quality. Both models saturate the L_∞ budget at ≈ 0.008 . Mean SNRs are 20.60 dB (Voxtral, $\sigma = 7.33$) and 16.43 dB (OpenS2S, $\sigma = 6.91$). On Voxtral, 55.0% of samples exceed 20 dB and 26.3% exceed 25 dB; on OpenS2S the distribution shifts lower, with 40.65% below 15 dB. The SNR difference reflects utterance characteristics rather than a deficiency in the attack: OpenS2S samples tend to be shorter and lower in energy, so the same absolute perturbation magnitude consumes a larger fraction of the signal power. The multi-resolution STFT loss (Section ??) mitigates perceptually concentrated distortions, but a formal listening study remains necessary to confirm subjective imperceptibility.

Per-emotion semantic patterns. Semantic preservation varies systematically across emotions on both models. Surprise achieves the highest preservation rates (Voxtral: 46.60%, OpenS2S: 23.23%), while sad records the lowest (Voxtral: 33.80%, OpenS2S: 12.94%). We attribute this asymmetry to utterance complexity: sad speech in ESD tends to be longer and more prosodically variable, giving the optimizer a larger acoustic surface to distort. The tension between emotion flipping and content preservation—especially pronounced

for sad speech—highlights a fundamental trade-off: the acoustic features that carry emotional coloring overlap substantially with those that encode lexical content, and the current loss weighting may not fully disentangle the two.

3.4 Summary

Three findings stand out from the white-box evaluation. First, the attack achieves high targeted success rates on both models (93.80% and 78.44%) under a perturbation budget that leaves traditional SER attacks essentially inert, confirming that ALLM-native attack design is both necessary and effective. Second, per-emotion patterns are consistent with the logit-level geometry characterized in Section ??: structurally anchored emotions resist perturbation, while concentrated clean-audio confidence provides a clear gradient handle. Third, semantic preservation remains the principal bottleneck, revealing a tension between emotion control and content fidelity that future work must address—whether through improved loss scheduling, content-aware perturbation constraints, or decoupled acoustic representations.

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